

Toheeb Eji

(647) 207-6299 | toheebaji@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

McMaster University | Hamilton, ON
B.S. in Computer Science - 3.78 GPA

Sep. 2025 - Present

PROJECTS

Sift | *Go, React, TypeScript, SQLite* [GitHub](#) Jul. 2026 - Aug. 2026

- Built a search engine in **Go** that crawls **500k pages** in ~30 min and indexes them in ~3 min, using a **concurrent crawler** with hostname-based rate limiting and dynamic worker pools.
- Cut a 20k-URL crawl from ~7 min to ~2 min (~**70%**) by redesigning the **crawl frontier** with a ready queue and cooldown **min-heap**, as well as adding a **DNS-failure negative cache**.
- Reduced crawler memory from 2 GB to ~**150 MB** on 20k pages by tuning HTTP connection pools.
- Improved result relevance and removed full-index loading at query time by building an **inverted index** with **BM25** ranking (title, domain, URL boosts) over **memory-mapped** binary posting files.

Serene | *React, Next.js, Typescript, Tailwind CSS, C#, ASP.NET Core* [GitHub](#) Jul. 2025 - Jan. 2026

- Shipped a full-stack mental health platform for McMaster students by building it with **Next.js**, **ASP.NET Core**, and **PostgreSQL**, deployed on **Vercel** and **Railway**.
- Personalized resource recommendations by user demographics and needs by building a custom **RAG** pipeline over a **vector database** with **OpenAI embeddings**.
- Automated context-aware prompt delivery by integrating **Google Gemini** and scheduling background jobs with **Quartz.NET**.

Diff Reviewer | *Python, LangChain, LangGraph, Pydantic* [GitHub](#) Sep. 2026 - Present

- Parallelized code review across **4 specialized agents** (bug, style, test, config safety) by building a **LangGraph** pipeline that parses git diffs, classifies changed files, and fans out in parallel
- Enabled **bug detection** beyond the diff by giving a **tool-calling LangChain** agent file-reading and code-search tools, returning findings as **Pydantic**-validated structured output.

HACKATHONS

Waypoint (DeltaHacks 12) | *TypesScript, Python, React, TailwindCSS, FastAPI, Mapbox GL JS* [GitHub](#) Jan. 2026

- Simulated **1000+ agents** for search-and-rescue planning by building a **FastAPI Monte Carlo** engine with terrain-aware movement from NASA SRTM elevation, weather **APIs**, and **OpenStreetMap**.
- Visualized predictive heatmaps and 12-hour forecasts across **500+** concurrent simulations in **real time** by building a **Mapbox GL JS** and **React** client with **TanStack Query**.
- Enabled auto-scaling and reliable production uptime by containerizing the server with **Docker** on **Google Cloud Run**.

StudyFlow (RythmHacks 2025) | *React Native, TypeScript, FastAPI, Groq, LangChain* [GitHub](#) Oct. 2025

- Winner of the **Best Venture Track Award**.
- Turned doom-scrolling into active-recall study breaks by building a **React Native (Expo)** app and **FastAPI** backend that uses **Groq LLMs** to convert uploaded PDFs into flashcards and multiple-choice questions.

TECHNICAL SKILLS

Languages: Go, C#, TypeScript, JavaScript, Python, SQL, HTML/CSS, Java, C/C++

Frameworks: React, Next.js, Svelte, React Native, ASP.NET Core, Entity Framework Core, FastAPI, Express.js, NestJS, Tailwind CSS, LangChain, LangGraph

Libraries: TanStack, Zod, Zustand, shadcn/ui, Mapbox GL, Playwright, xUnit, React Testing Library, pgvector

Developer Tools: Git, GitHub Actions, Docker, PostgreSQL, SQLite, Redis, Vercel